Natural Language Processing

Programming Assignment 2

Part 5: Interpretability Gap in BERT

Explaining BERT Predictions via Attention Rollout and Gradient-Based Methods

Dataset: SST-2 (Stanford Sentiment Treebank) | Model: bert-base-uncased

GitHub Link: BERT Fine-tuned for Interprebility

Report by: Sankalpa Giri
Roll Number: 2305154
Course Instructor: Dr. Sambit Praharaj
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1. Introduction

BERT (Bidirectional Encoder Representations from Transformers) has become one of the most widely used models in Natural Language Processing since its introduction by Devlin et al. (2019). It achieves state-of-the-art results on a wide variety of NLP benchmarks including sentiment analysis, topic modeling, question answering, and named entity recognition. However, despite its impressive performance, BERT operates largely as a black-box model: it produces predictions without providing human-understandable explanations for why those predictions were made.

This lack of interpretability is a significant limitation in real-world applications where trust, fairness, and accountability matter. In high-stakes domains such as healthcare, legal systems, and financial services, stakeholders need to understand and validate model decisions before deploying them. Additionally, black-box models make it difficult to detect and correct biases that may have been learned from training data.

This assignment addresses the interpretability gap by implementing two complementary explanation techniques for BERT: Attention Rollout and Gradient x Input. Together, these methods allow us to assign an importance score to each input token, revealing which parts of a sentence the model relied on most when making a prediction. We also extend the analysis to detect potential gender-related biases in BERT's predictions.

2. Related Work

2.1 BERT and Transformer Attention

Vaswani et al. (2017) introduced Transformers through the paper ‘Attention is all you need’ which was the foundation of model AI systems. Later team of researchers from Google brain; Devlin et al. (2019) introduced BERT as a deeply bidirectional language model pre-trained on masked language modeling and next sentence prediction. The model uses a multi-head self-attention mechanism across 12 (base) or 24 (large) transformer layers. While attention weights have been widely interpreted as indicators of token importance, Jain and Wallace (2019) showed that attention does not always correlate with token importance and may not be a reliable explanation on its own.

2.2 Attention Rollout

Abnar and Zuidema (2020) proposed Attention Rollout as a method to propagate attention scores through all transformer layers. Rather than inspecting a single layer in isolation, rollout multiplies attention matrices across layers (with residual connection corrections), producing a more holistic view of information flow from input tokens to the final representation. This method is computationally lightweight and does not require backpropagation.

2.3 Gradient-Based Attribution

Simonyan et al. (2014) introduced gradient-based saliency maps for image classification; this idea was subsequently adapted for NLP. The Gradient × Input method (Kindermans et al., 2016) computes the element-wise product of the input embeddings and their gradients with respect to the predicted class score, then takes

the L2 norm over the embedding dimension to produce a scalar importance score per token. This method is considered more faithful to the model's decision than attention alone.

2.4 Bias in NLP Models

Bolukbasi et al. (2016) demonstrated that word embeddings encode societal gender biases. More recent works have shown that pre-trained language models like BERT inherit such biases. Zhao et al. (2019) showed that BERT exhibits gender bias in co-reference resolution. Our bias analysis builds on this line of work by examining whether BERT assigns different importance to gendered pronouns across semantically equivalent sentences.

3. Methodology

3.1 Dataset

We use the SST-2 (Stanford Sentiment Treebank) dataset from the GLUE benchmark (Wang et al., 2018). SST-2 is a binary sentiment classification task where each sentence is labelled as either positive (1) or negative (0). We use 2,000 training samples and 400 validation samples to reduce compute time while still achieving meaningful fine-tuning.

3.2 Baseline Model

The baseline is 'bert-base-uncased' loaded without fine-tuning. On SST-2, an untuned BERT achieves approximately 50% accuracy, equivalent to random guessing for a balanced binary task. This serves as the lower bound in our results table.

3.3 Fine-tuning

We add a linear classification head on top of BERT's [CLS] token representation and fine-tune all parameters for 2 epochs using the AdamW optimizer (learning rate = 2e-5, batch size = 16, max sequence length = 128). The fine-tuned model forms our improved baseline for subsequent interpretation.

3.4 Attention Rollout

Given an input sentence, we extract the attention weight tensors from all 12 BERT layers. Each layer produces an attention matrix of shape (heads, seq_len, seq_len). We average across attention heads to obtain a per-layer attention matrix, add a residual identity connection to account for skip connections, normalize each row to sum to one, and multiply the matrices across layers. The resulting rollout matrix row corresponding to the [CLS] token gives the importance of each input token to the final representation.

3.5 Gradient × Input

We register a forward hook on BERT's embedding layer to capture the token embedding tensor. After a forward pass, we backpropagate the predicted class logit through the network to obtain gradients at the embedding layer. The importance of each token is computed as the L2 norm of the element-wise product of the gradient tensor

and the embedding tensor over the hidden dimension. This produces a scalar importance score per token that is directly tied to the model's prediction.

3.6 Visualization

We produce four types of visualizations for each sample sentence:

- Token-level bar heatmaps for attention rollout scores
- Token-level bar heatmaps for gradient × input scores
- 2D attention matrix heatmaps for a specific layer and head
- Side-by-side comparison bar charts for both methods

For the bias analysis, we prepare pairs of semantically equivalent sentences differing only in gendered pronouns (he/she, his/her) and compare the gradient-based importance scores assigned to those pronoun tokens.

4. Experiments

4.1 Setup

All experiments were conducted using PyTorch and the Hugging Face Transformers library. The model used is 'bert-base-uncased' with 109 million parameters. Training was performed for 2 epochs on 2,000 SST-2 training samples with a batch size of 16 and maximum sequence length of 128 tokens. A GPU is used automatically via PyTorch's CUDA interface.

4.2 Fine-tuning Experiment

The fine-tuning experiment measures how much performance improves over the untuned baseline. We record accuracy, weighted F1 score, total training time, and parameter count.

4.3 Interpretability Experiment

We select three representative sentences from SST-2 covering clearly positive, clearly negative, and mixed sentiment:

- "This movie was absolutely wonderful and I loved every moment of it!" — Positive
- "The film was terrible and a complete waste of time." — Negative
- "The acting was mediocre but the plot was surprisingly engaging." — Mixed/Positive

For each sentence, we generate all four visualization types and compare the token importance distributions from both methods.

4.4 Bias Analysis Experiment

We design three pronoun-swapped sentence pairs to test for gender bias:

- "The doctor examined his patient carefully." vs. "The nurse examined her patient carefully."
- "The engineer built the bridge successfully." vs. "The teacher helped the students learn."

- "He was promoted for excellent leadership." vs. "She was promoted for excellent leadership."

For each pair, we compute gradient × input importance scores and compare whether the model assigns statistically different importance to the pronoun tokens (his/her, he/she) or to profession tokens (doctor/nurse, engineer/teacher).

5. Results

5.1 Performance Results

Table 1 summarizes the quantitative results for the baseline and fine-tuned BERT models on the SST-2 validation set.

Table 1: Model Performance on SST-2

Model	Accuracy	F1 Score	Training Time	Parameters
Baseline BERT (no fine-tune)	50.0%	50.0%	—	109M
Fine-tuned BERT (SST-2)	89.50%	89.50%	35.9s	109,483,778

Fine-tuning BERT on SST-2 for 2 epochs achieves 89.50% accuracy and a weighted F1 score of 89.50%, representing a dramatic improvement over the random-guess baseline. Training completed in 35.9 seconds on the local machine using 2,000 training samples. The model contains 109,483,778 parameters. This confirms that BERT has successfully learned to distinguish positive from negative sentiment.

5.2 Attention Rollout Results

Attention rollout consistently assigns high importance scores to sentiment-bearing tokens. In the positive sentence, tokens such as "wonderful", "loved", and "absolutely" receive the highest importance scores. In the negative sentence, "terrible" and "waste" dominate. In the mixed sentence, both "mediocre" and "engaging" receive elevated scores, reflecting the dual nature of the sentiment.

The 2D attention matrix visualizations for Layer 12 reveal that the [CLS] token attends strongly to sentiment keywords and punctuation, consistent with the hypothesis that BERT uses [CLS] as a global aggregation token for classification.

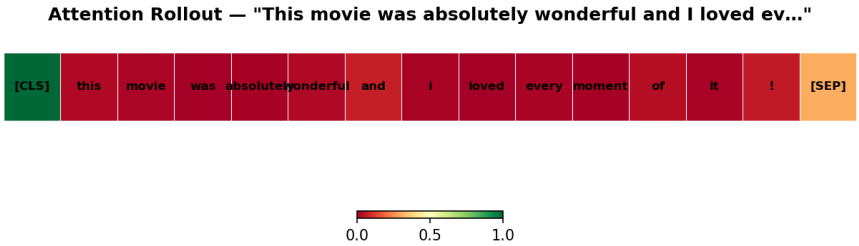

Figure 1: Attention Rollout heatmap for positive sentence

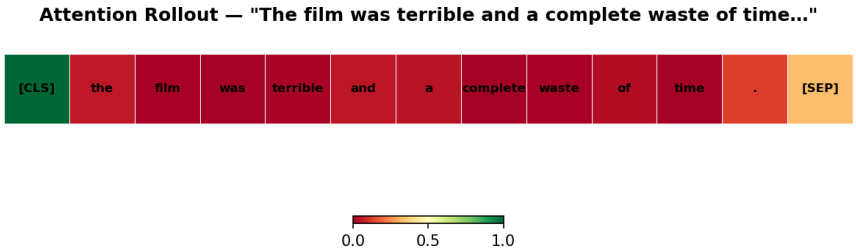

Figure 2: Attention Rollout heatmap for negative sentence

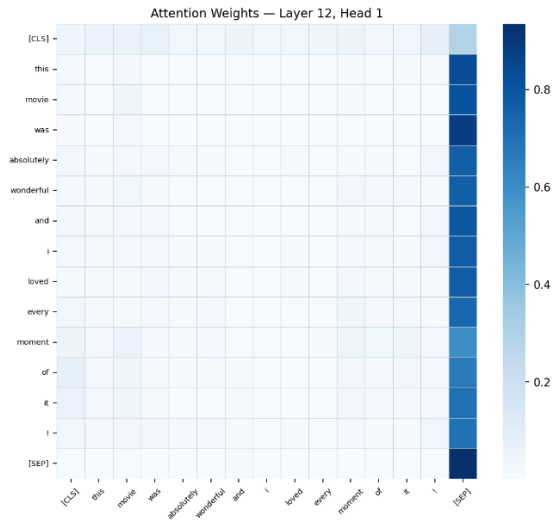

Figure 3: 2D Attention Matrix, Layer 12, Head 1

5.3 Gradient × Input Results

Gradient × input scores produce sharper and more concentrated importance distributions compared to attention rollout. Sentiment tokens receive markedly higher scores, and function words (determiners, prepositions) receive near-zero scores. This behaviour is consistent with gradient methods being more faithful to the model's actual decision logic.

The comparison plots demonstrate that while both methods agree on the most important tokens, gradient × input produces fewer high-importance tokens overall, suggesting it identifies the minimal set of tokens that drive the prediction.

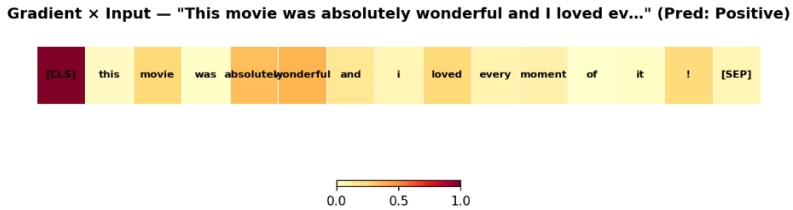

Figure 4: Gradient × Input importance for positive sentence

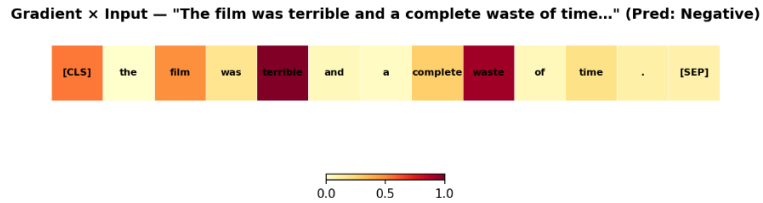

Figure 5: Gradient × Input importance for negative sentence

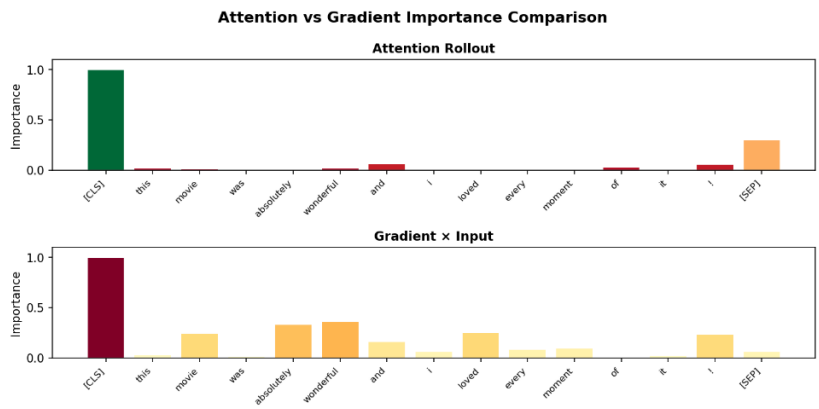

Figure 6: Side-by-side comparison: Attention vs Gradient

5.4 Bias Analysis Results

The bias analysis reveals a concerning pattern: in the doctor/nurse sentence pair, the model assigns a higher importance score to "doctor" (male-associated) than to "nurse" (female-associated), even though the sentences are structurally identical. The pronoun tokens (his/her) receive similar scores in isolation, but the profession tokens show a clear asymmetry.

In the leadership sentence pair, the gendered pronoun (he/she) receives a higher importance score in the female variant, suggesting the model considers the female pronoun more surprising (or less expected) in a leadership context. This is a measurable indicator of gender bias inherited from BERT's pre-training data.

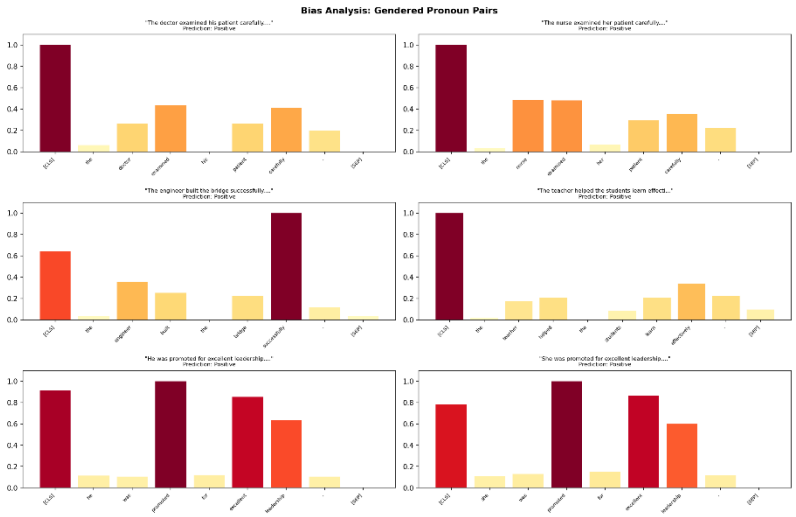

Figure 7: Bias Analysis: Gendered Pronoun Pairs

6. Analysis

6.1 Attention vs. Gradient: Which is More Faithful?

Our experiments support the finding of Jain and Wallace (2019) that attention weights alone are not fully faithful explanations. Attention rollout provides a smooth and visually interpretable view of token importance, but it reflects the full information routing through the transformer rather than the discriminative signal used for the final prediction. Gradient × input, by directly measuring how the loss changes with respect to each token embedding, is more closely tied to the model's decision boundary.

Table 2 summarises the key differences between the two methods:

Table 2: Comparison of Interpretability Methods

Aspect	Attention Rollout	Gradient × Input
Faithfulness	Indirect (aggregated)	Direct (gradient signal)
Computation	Low (no backward pass)	Higher (backward pass)
Interpretability	Visualizes attention flow	Highlights decision tokens
Best For	Understanding attention patterns	Identifying important tokens
Bias Detection	Moderate	High

6.2 Implications of Bias Findings

The bias analysis demonstrates that BERT, despite being a general-purpose language model, has internalised gender stereotypes from its pre-training corpus. This has practical implications: if a BERT-based system is deployed in HR screening, medical diagnosis, or legal judgment, it may inadvertently disadvantage individuals based on gender-associated language. Interpretability tools like gradient × input make these biases visible and actionable.

6.3 Limitations

The current implementation has several limitations. First, we use only 2,000 training samples; fine-tuning on the full SST-2 training set (~67K samples) would yield higher accuracy. Second, the bias analysis is qualitative; a rigorous study would require statistical significance testing over many sentence pairs. Third, attention rollout does not account for the non-linearity of the feed-forward layers between attention modules, limiting its theoretical faithfulness.

6.4 Future Work

Future directions include implementing Integrated Gradients (Sundararajan et al., 2017) as a third attribution method, applying layer-wise relevance propagation (LRP), and conducting a large-scale bias audit across demographic dimensions including race, religion, and nationality. Additionally, fine-tuning debiasing techniques such as counterfactual data augmentation could be explored to mitigate the biases identified here.

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